

PSLE MATHEMATICS PRACTICE PAPER 4

Speed, Distance & Time

Duration: 1 hour

Total Marks: 50

Name: _____

Date: _____

SECTION A: Short Answer Questions (20 marks)

1. A car travels 150 km in 3 hours. Find its average speed. (2 marks)

2. A runner runs at 8 m/s. How far does she run in 10 seconds? (2 marks)

3. A train travels at 60 km/h for 2.5 hours. Find the distance. (2 marks)

4. A cyclist covers 45 km in 1.5 hours. Find his speed. (2 marks)

5. An object moves at 20 m/s. How long does it take to cover 500 m? (2 marks)

6. A plane flies 1200 km at 400 km/h. How long is the flight? (2 marks)

7. Convert 72 km/h to m/s. (2 marks)

8. Convert 15 m/s to km/h. (3 marks)

SECTION B: Long Answer Questions (30 marks)

9. John jogs at 5 m/s for 12 seconds, then walks at 1.5 m/s for 20 seconds. Find the total distance covered. (4 marks)

10. A bus travels 120 km in 2 hours in the morning and 180 km in 1.5 hours in the afternoon. Find the average speed for the whole journey. (4 marks)

11. Sarah walks at 1.2 m/s. How long will it take her to walk 1.8 km? (3 marks)

12. A car travels at 80 km/h for the first 240 km, then at 60 km/h for the next 120 km. Find the average speed for the entire journey. (4 marks)

13. Two cyclists start from the same point. One travels at 12 km/h and the other at 15 km/h. How far apart will they be after 30 minutes? (3 marks)

14. An athlete runs 400 m in 50 seconds. Find his speed in m/s and km/h. (4 marks)

15. A car travels at 90 km/h. If it stops for 30 minutes in the middle, and the total journey takes 3.5 hours, what is the total distance? (4 marks)

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SOLUTIONS

SECTION A SOLUTIONS

Q1: Average speed

$$\text{Speed} = \text{Distance} \div \text{Time} = 150 \div 3 = 50 \text{ km/h}$$

Q2: Distance from speed

$$\text{Distance} = \text{Speed} \times \text{Time} = 8 \times 10 = 80 \text{ m}$$

Q3: Distance

$$\text{Distance} = \text{Speed} \times \text{Time} = 60 \times 2.5 = 150 \text{ km}$$

Q4: Speed

$$\text{Speed} = \text{Distance} \div \text{Time} = 45 \div 1.5 = 30 \text{ km/h}$$

Q5: Time

$$\text{Time} = \text{Distance} \div \text{Speed} = 500 \div 20 = 25 \text{ seconds}$$

Q6: Time

$$\text{Time} = \text{Distance} \div \text{Speed} = 1200 \div 400 = 3 \text{ hours}$$

Q7: Convert 72 km/h to m/s

$$72 \text{ km/h} = 72 \times 1000 \div 3600 = 20 \text{ m/s}$$

Q8: Convert 15 m/s to km/h

$$15 \text{ m/s} = 15 \times 3600 \div 1000 = 54 \text{ km/h}$$

SECTION B SOLUTIONS

Q9: Total distance

$$\text{Distance jogging} = 5 \times 12 = 60 \text{ m, Distance walking} = 1.5 \times 20 = 30 \text{ m, Total} = 90 \text{ m}$$

Q10: Average speed

$$\text{Total distance} = 120 + 80 = 200 \text{ km, Total time} = 2 + 1.5 = 3.5 \text{ hours, Average} = 200 \div 3.5 = 57.14 \text{ km/h}$$

Q13: Distance apart

$$\text{First cyclist: } 12 \times 0.5 = 6 \text{ km, Second cyclist: } 15 \times 0.5 = 7.5 \text{ km, Difference} = 1.5 \text{ km}$$

Q14: Speed conversions

Speed = $400 \div 50 = 8 \text{ m/s}$, In km/h = $8 \times 3.6 = 28.8 \text{ km/h}$

Q12: Average speed

Total distance = $240 + 120 = 360 \text{ km}$, Total time = $240 \div 80 + 120 \div 60 = 3 + 2 = 5 \text{ hours}$,
Average = $360 \div 5 = 72 \text{ km/h}$

Q15: Distance

Travel time = $3.5 - 0.5 = 3 \text{ hours}$ (excluding stop), Distance = $90 \times 3 = 270 \text{ km}$

Q11: Time

Time = Distance $\div$ Speed = $1800 \text{ m} \div 1.2 \text{ m/s} = 1500 \text{ seconds} = 25 \text{ minutes}$

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